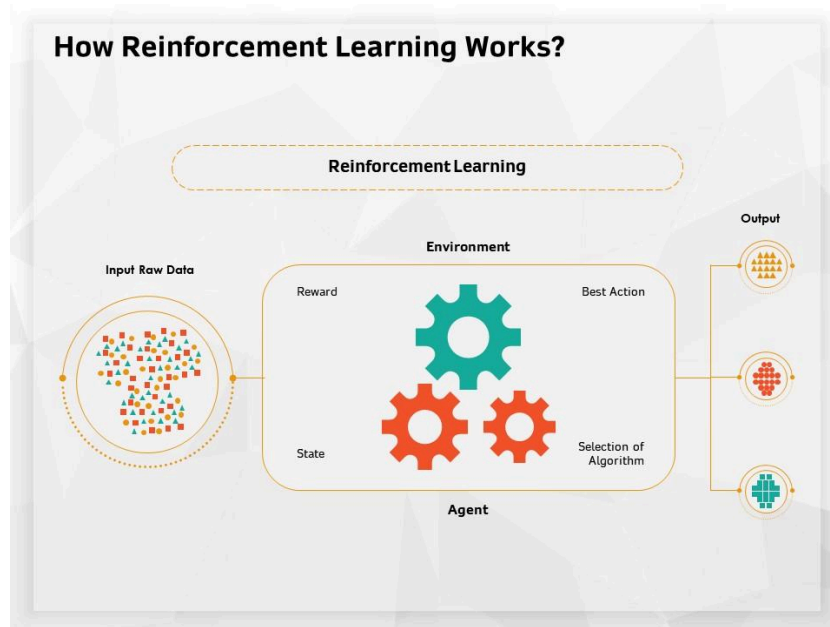


behavior; instead, the agent receives feedback in the form of **rewards** for its actions and seeks to maximize the cumulative reward over time. In a typical reinforcement learning scenario, an *agent* observes the state of the environment, takes an *action*, and then receives a *reward* and a new *state* from the environment in response. Over many cycles of interaction, the agent's goal is to learn a policy (a strategy of choosing actions) that yields the highest long-term reward. This trial-and-error process is what allows the agent to improve its performance with experience.



Source: slideteam.net

Project Ideas for Reinforcement Learning

- **Game Playing AI:** Develop an RL agent to play a game (such as Pac-Man, Chess, or an Atari game). The agent will learn strategies through trial and error by earning points or winning/losing games as reward feedback, eventually becoming proficient at the game.
- **Autonomous Driving Simulator:** Train a self-driving car agent in a simulation environment. The agent receives positive rewards for safe driving (staying in lane, avoiding collisions, reaching destinations) and negative rewards for crashes or heavy swerving, learning to drive safely over many simulated miles.
- **Robot Arm Control:** Use reinforcement learning to teach a robotic arm to accomplish a task, like picking up and sorting objects. The robot is rewarded for successfully grasping and placing items correctly, encouraging it to refine its motions to complete the task more reliably.